

Your LAST Name _____

Your FIRST Name _____

Your Collaborators

• Name: _____ SID: _____

• Name: _____ SID: _____

• Name: _____ SID: _____

• Name: _____ SID: _____

Policy Statement

- We encourage you to collaborate, but only in a group of up to *five* current EE 120 students.
- Be sure to write the names of your collaborators at the appropriate location on the next page. Each collaborator must submit a distinct, self-prepared solution document containing original contributions to the collaborative effort.
- To submit your work, you have several options:
 - (a) Print out this homework document and write your solutions on the printout. Then upload a *PDF file* of the scan.
 - (b) Write your solutions on this PDF file using a tablet device. Then upload a *PDF file*.
 - (c) Use your own sheets of paper, but demarcate clearly the space for each problem, part, and subpart. Then upload a *PDF file*.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it*. Remember that, even though the problem sets are self-graded, we will evaluate your work whenever we check your self grades.
- Unless we explicitly state otherwise, you may expect full credit *only if* you explain your work succinctly, but clearly and convincingly.
- If you are asked to provide a “sketch,” it refers to a well-labeled *hand-drawn* sketch—not a plot generated by a computing device.
- On occasion, a problem set contains one or more problems designated as “optional.” Your solutions to such problems will NOT be evaluated. Nevertheless, you’re responsible for learning the subject matter within their scope.
- If you have a confirmed disability that precludes you from complying fully with these instructions or with any other parameter associated with this problem set, please alert us immediately about reasonable accommodations afforded to you by the DSP Office on campus.

Overview

This problem set explores

- DT State Space
- The Dirac delta

HW5.1 (A Jolly Famous Sequence of Numbers)

The input x and the output y of a causal, discrete-time LTI system H are governed by the second-order linear constant-coefficient difference equation

$$y[n] = y[n-1] + y[n-2] + x[n],$$

where the system is assumed to be initially at rest—that is, $y[n] = 0$ for all $n < 0$.

- (a) Determine $h[n]$, the impulse response values of the system H . What famous sequence of numbers does the impulse response h trace? Is the system BIBO stable? Provide a succinct, yet clear and convincing explanation.

- (b) Draw a delay-adder-gain (DAG) block diagram implementation of the system.

- (c) On the DAG block diagram implementation of the previous part, label the appropriate state variables $q_1[n]$ and $q_2[n]$. Then proceed to write the corresponding state-space equations

$$\mathbf{q}[n+1] = \mathbf{A} \mathbf{q}[n] + \mathbf{B} x[n] \quad \text{State-Evolution Equation}$$

and

$$y[n] = \mathbf{C} \mathbf{q}[n] + \mathbf{D} x[n], \quad \text{Output Equation}$$

where $\mathbf{q}[n] = [q_1[n] \ q_2[n]]^T$ is the state vector at time n . You must, of course, determine the matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ completely.

- (d) Determine the modes $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ of the system. For consistency, show that the modes are of the form $\left(\lambda_1, \mathbf{v}_1 = \begin{bmatrix} 1 \\ -\lambda_2 \end{bmatrix}\right)$ and $\left(\lambda_2, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -\lambda_1 \end{bmatrix}\right)$, where λ_1 and λ_2 are the eigenvalues that you must determine. Also, explain why the eigenvectors *must* be mutually orthogonal.

- (e) Is the system *internally stable*? Provide a succinct, yet clear and convincing explanation.

HW5.2 (Dirac Delta Properties)

- (a) (**Sampling Property**) Simplify, and provide a well-labeled sketch of, each of the following expressions involving the Dirac delta:

(i) $(t^2 + 1) \delta(t)$

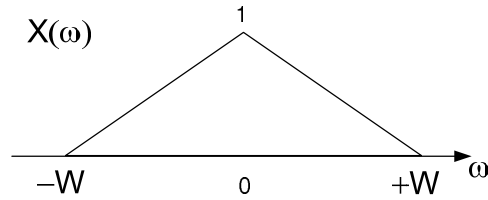
(ii) $e^{\alpha t} \delta(t - 1)$, where $\alpha \in \mathbb{C}$

(iii) $\cos\left(\pi t + \frac{\pi}{2}\right) \delta\left(t - \frac{1}{2}\right)$

(iv) $X(\omega) P(\omega)$, where P is the impulse train

$$P(\omega) = \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{kW}{2}\right),$$

with $W > 0$, and $X(\omega)$ is as shown below:



(b) **(Sifting Property)** Evaluate each of the following expressions involving the Dirac delta:

(i) $\int_{-\infty}^{+\infty} e^{\alpha t} \delta(t - 1) dt$, where $\alpha \in \mathbb{C}$

$$(ii) \int_{0.49}^{0.51} \sin(\pi t) \delta\left(t - \frac{1}{2}\right) dt$$

$$(iii) \int_{-\infty}^{+\infty} X(\omega) \delta\left(\omega - \frac{2W}{3}\right) d\omega, \text{ where } X(\omega) \text{ is as shown in part (a)(iv) above.}$$

$$(iv) \int_{-\infty}^{+\infty} e^{-|t|} v(t) dt, \text{ where } v \text{ is the impulse train}$$

$$v(t) = \sum_{k=-\infty}^{+\infty} \delta(t - k).$$

HW5.3 (Dilation, Contraction, and Time-Reversal Properties)

Show that scaling the argument of the Dirac delta affects its strength (area) in the following manner:

$$\delta(at) = \frac{1}{|a|} \delta(t), \quad \forall a \in \mathbb{R}. \quad (1)$$

Equation (1)—which denotes an operational equivalence, not a point-wise equality—is sometimes referred to as the *time-scaling* property, even though the argument of the Dirac delta need not be “time.”

To verify the time-scaling property, place each side of Equation (1) inside an integral with a test function g that is continuous at $t = 0$, and verify that both sides produce the same result. That is, show that

$$\int_{-\infty}^{+\infty} \delta(at) g(t) dt = \int_{-\infty}^{+\infty} \frac{1}{|a|} \delta(t) g(t) dt. \quad (2)$$

Hint: Use the change of variable $\tau = at$ in the left-hand side of Equation (2).

Show that the Dirac delta *behaves like* an even function, namely, $\delta(-t) = \delta(t)$.

Note that this is true even if the function that the Dirac delta idealizes is not even.